

What is the Impact of Inflation on Consumption in Canada?

Final Paper

Presented by:

Carter Goveas

For:

Honours Capstone Seminar

ECON 4905B

April 8, 2025

Abstract:

This research project aims to analyse the impact of inflation on consumers behaviour in Canada. There are four models that have been used in order to answer this question. The first model analyses the impact of the Consumer Price Index (CPI) on consumer spending. The second model analyses the impact of the Gross Domestic Product (GDP) deflator on consumer spending. The third model analyses the impact of the CPI trim on consumer spending. The fourth and final model analyses the impact of CPI median on consumer spending. The hypothesis for each model was that as inflation increases, so would consumer spending. The results supplement this hypothesis for each model.

Introduction

Inflation is a very complex concept that is extremely difficult to fully understand. It has many causes and implications which are still being researched to this day. There are many compelling questions that are always asked about inflation. What causes inflation? Why does inflation increase? Why does inflation decrease? Is there a way to avoid inflation? This research attempts to answer another difficult question which is: How does inflation affect consumer spending? More specifically, this research is being conducted strictly in Canada, across a 20 year period between the years of 2004 and 2023. There are a few potential outcomes on consumer spending that can transpire as a result of inflation. First, consumers could decide that they want to spend more money. This could happen since “Elevated inflation discourages saving because it erodes the purchasing power of savings over time. That prospect can encourage consumers to spend and businesses to invest.” (Floyd, 2024) Second, consumers could decide that they want to spend the same amount of money, causing there to be no difference before and after inflation. This could be due to the fact that some people think “...inflation will remain low in the long run...” so no real change is needed to their consumption habits. (Bank of Canada, 2020) Finally, consumers could decide that they would like to spend less money. This could be due to the notion that an increase in inflation will decrease a consumers’ purchasing power, driving them to buy less. (Scotiabank, n.d.) The hypothesis leading this research is that as a result of inflation increasing, consumer spending will increase subsequently.

In order to conduct this research, four different models have been used. Each model regresses consumer spending on a different inflation indicator. The inflation indicators are CPI, GDP deflator, CPI trim, and CPI median. For CPI and GDP deflator, Cochrane-Orcutt Regression were used. For CPI trim and CPI median, OLS regressions were used. In order to ensure accurate and well-rounded models, control variables were also added. These variables

consist of GDP growth and interest rate. Both of these control variables affect consumer spending, therefore adding them to the models makes the results better explained for.

Additionally, an interaction variable is also added to all four models, specifically a time trend variable. This helps account for regression problems which will be touched on later in this research.

Not only does this research try to discover the impact of inflation on consumer spending, but it also attempts to answer other things such as the extent to which inflation affects consumer spending. By using four different models to represent inflation, this research also attempts to uncover which indicator of inflation is most accurate and provides the most meaningful results. This research answers all the questions it wishes to, uncovering useful information which can be used for future research/analyses’.

Section 1: Literature Review

1. The Impact of Inflation Rate on Private Consumption Expenditure and Economic Growth - Evidence from Ghana by Babalol Emmanuel Olusola, Murako Emmanuel Chimezie, Selma Magano Shuuya, and George Yaw Asare Addeh (2022)

This piece of literature uncovers how inflation affects private consumption expenditure and economic growth in Ghana. This research was conducted between the years of 1964 to 2013. It was concluded that there is a negative relationship between inflation and private consumption expenditure. This result was found by using a variety of different frameworks and theories. The monetarist approach, the structural theory of inflation, and the life cycle theory of consumption are just some of the theories that are used in this piece of literature. In terms of methodology, there were different techniques used such as the Pearson Product Moment Correlation, the Unit

Root Test, the Engle-Granger Co-Integration Test, Ordinary Least Squares (OLS), and the Granger Causality Test. The main technique that I drew from this piece of literature is that of OLS. It was used by Olusola et al. in order “...to look at how inflation impacts consumption expenditure over time.” (cite) I used this technique for the same reason and it proved to be very effective. In order to properly implement OLS, Olusola et al. regressed private consumption expenditure (dependent variable) on inflation rate (independent variable), interest rate, and gross domestic product growth rate (control variables). I used a very similar approach by also using an inflation indicator as my independent variable and using GDP growth and interest rate as control variables in each of my models. This piece of literature has been extremely helpful in the making, and results, of my research.

2. What is the effect of inflation on consumer spending behaviour in Ghana? By

Gabriel Effah Nyamwekye and Eugene Adusei Poku (2017)

This piece of literature also reveals the impact of inflation on consumer spending behaviour in Ghana. It uses an identical time frame from 1964 to 2013 with annual data. To conduct this research, Effah & Poku use three different procedures. They use the Ordinary Least Squares test (OLS), the Johansen test (JH), and the Vector Error Correction test (VECM). Each test provided substantial results. The OLS test signified a positive relationship between inflation and consumer spending. The JH test signified a strong long run relationship between inflation and consumer spending. The VECM test showed a strong short run relationship between inflation and consumer spending. I created my hypothesis under the guise of these results, as well as general results in research done. This hypothesis being that there is a positive relationship between inflation and consumer spending. I also used identical measures for inflation and

consumer spending as this measure. Effah & Poku used CPI to measure inflation, and household consumption expenditure to measure consumer spending. They got this data from the World Bank. This piece of literature has been super useful in the making, and results, of my research.

Section 2: Data Description

For this research, I use a variety of different data sources in order to fully analyse the research question at hand. I found that using different data sources helped create a more complete and well rounded analysis.

For starters, there is the data used for consumer spending. I obtained this data from Statistics Canada. The data was originally given as a quarterly statement of Canadian Final Household Expenditure. The data was given from 2004 up to 2023. I had to manually clean this data so that I could get an annual Canadian Final Household Expenditure rather than quarterly since my analysis runs on an annual basis. I chose Canadian Final Household Expenditure to represent my consumer spending variable because this was the most commonly used/researched way to describe consumer spending, based on my findings.

Next, there is the data that I am using for CPI. I also gathered this data from Statistics Canada. The data was given annually, not seasonally adjusted, making my cleaning of the data a lot easier. I chose to use CPI as an inflation indicator since it is one of the most common variables used to explain inflation. The research done on it is very thorough and has been happening for a while which makes it reliable, especially from a source such as Statistics Canada.

The data used for the GDP deflator was gathered from the World Bank. This data was given on an annual basis and was very easy to clean. The world bank defines GDP deflator as

“...the ratio of GDP in current local currency to GDP in constant local currency.” This is a very fit indicator to help measure inflation as it is one of the main indicators that is trusted and researched. Therefore it gave strength to my results in this paper.

The next set of data used is for GDP growth. For this, I obtained the data from the World Bank. The data was given as the annual percentage change in GDP growth. I used this data for GDP growth as a control variable to help better explain the models that I am regressing on. Annual percentage change is a very concrete and reliable measure for GDP growth.

Next, is the data for the interest rate. I gathered this data from the Federal Reserve Bank of St. Louis. The data is better defined as the “Long-Term Government Bond Yields: 10-Year”. It is specifically for Canada, despite the data coming from an American reserve bank. This data is also used for the interest rate as a control variable. “A bond yield is the return an investor [makes] on a bond”, which makes it a very suitable measure for interest rates in Canada in my models. (Hayes, 2024)

Finally, for both my CPI trim and CPI median, I used the Bank of Canada to gather my data. These measures for CPI differ from standard CPI and act as core measures of inflation. CPI trim is based on the trimmed-mean of CPI, and CPI median is based on the weighted mean of CPI. () Both these indicators offer different outlooks on CPI and inflation which is useful in this paper in order to get a better grasp on how consumer spending is correlated/affected by inflation.

In total, I obtained seven different datasets across all four models. Within these seven datasets, they come from four different sources. I find that using a variety of sources will reduce bias and help diversify the research. All of the data used has been a key component in uncovering the impacts of inflation on consumer spending.

Section 3: Methodology

This research project was completed using Stata and different regression techniques in order to analyse the effects of inflation on consumer spending. I used four different models and two different regression techniques during my analysis. The four models include regressing consumer spending on four different inflation indicators which include CPI, GDP deflator, CPI trim, and CPI median. The two regression techniques I used are pooled OLS and Cochrane-Orcutt.

OLS “...is a common technique for estimating coefficients of linear regression equations which describe the relationship between one or more independent quantitative variables and a dependent variable...” (Lumivero, n.d.-b) I found that OLS would be very beneficial for my research which is why I used it for two out of four of my models.

Cochrane-Orcutt “...is a well-known econometric approach to take serial correlation into account in the error term of linear model[s]. In [the] case of serial correlation, the usual linear regression method is invalid because the standard errors are not unbiased.” (Lumivero, n.d.-a) I used this estimation technique since my data and regressions proved to be autocorrelated. The Cochrane-Orcutt regression helped two out of four of my models and ensured accurate and strong results.

Model 1: Regressing Consumer Spending (CS) on Consumer Price Index (cpi) and Gross Domestic Product Growth (gdpgrowth) and Interest Rate (interestrates) and a time trend variable (time)

$$CS = \beta_0 + \beta_1 cpi + \beta_2 gdpgrowth + \beta_3 interestrates + \beta_4 time + \varepsilon$$

For this regression, I am using a Cochrane-Orcutt regression. I also implement differencing into my CS, cpi, gdpgrowth, and interestrate variables in order to fix for autocorrelation.

Additionally, I implement a time trend variable for the same reason which helps track how much the independent variable is explained by time. (citation)

Model 2: Regressing Consumer Spending (CS) on Gross Domestic Product Deflator (deflator) and Gross Domestic Product Growth (gdpgrowth) and Interest Rate (interestrate) and a time trend variable (time)

$$CS = \beta_0 + \beta_1 deflator + \beta_2 gdpgrowth + \beta_3 interestrate + \beta_4 time + \varepsilon$$

For this regression, I am using a Cochrane-Orcutt regression. I similarly implement differencing and a time trend variable to help fix for autocorrelation.

Model 3: Regressing Consumer Spending (CS) on Consumer Price Index Trim (trim) and GDP Growth (gdpgrowth) and Interest Rate (interestrate) and a time trend variable (time)

$$CS = \beta_0 + \beta_1 trim + \beta_2 gdpgrowth + \beta_3 interestrate + \beta_4 time + \varepsilon$$

For this regression, I am using pooled OLS. I use differencing for my variables to help account for autocorrelation.

Model 4: Regressing Consumer Spending (CS) on Consumer Price Index Median (median) and GDP Growth (gdpgrowth) and Interest Rate (interestrate) and a time trend variable (time)

$$CS = \beta_0 + \beta_1 median + \beta_2 gdpgrowth + \beta_3 interestrate + \beta_4 time + \varepsilon$$

For this regression, I am using pooled OLS. I use differencing and a time trend variable to help fix for autocorrelation in the model.

Predicted Impact On Consumer Spending:

Independent Variables

Consumer Price Index $\rightarrow (+)$

Gross Domestic Product Deflator $\rightarrow (+)$

Consumer Price Index Trim $\rightarrow (+)$

Consumer Price Index Median $\rightarrow (+)$

Control Variables

Gross Domestic Product Growth $\rightarrow (+)$

Interest Rate $\rightarrow (+)$

Interaction Variables

Time $\rightarrow (+)$

Section 4: Results

Model #1: Cochrane-Orcutt Regression - Regressing Consumer Spending on CPI

Iteration 0: $\rho = 0.0000$

Iteration 1: $\rho = -0.0242$

Iteration 2: $\rho = -0.0257$

Cochrane-Orcutt regression

CS	Coefficient	Std. err.	t	P>t	[95% confidence interval]
cpi	14478.200	3263.444	4.440	0.000	7592.936 21363.470
gdpgrowth	4109.700	1375.053	2.990	0.008	1208.592 7010.809
interestrate	15969.800	10931.730	1.460	0.162	-7094.127 39033.720
time	-1362.917	991.744	-1.370	0.187	-3455.314 729.479
_inter	2751898	1994286	1.380	0.185	-1455678 6959475
rho	-0.026	0.242	-0.110	0.915	-0.537 0.485

Number of Obs.	18
R-squared	0.8678
F(4, 13)	21.33
Prob > F	0.0000

Durbin-Watson statistic (original) = 2.040845

Durbin-Watson statistic (transformed) = 2.002354

The regression of consumer spending on CPI has an r-squared of 0.8678, implying 86.78% of the variance in the dependent variable in this model is explained by the independent variables. The F-statistic in this model is 0.0000 indicating overall significance.

The p-value for CPI is 0.000, which indicates an extremely good significance level. CPI has a positive coefficient of 14478.200 which suggests that as CPI increases, so does consumer spending. This reflects what I hypothesized.

The p-value for GDP growth is 0.008, which suggests an extremely good significance level. GDP growth has a positive coefficient of 4109.700 implying an increase in GDP growth would mean an increase in consumer spending. This aligns with what was hypothesized.

The p-value for interest rate is 0.162 which suggests insignificance. Interest rate has a

positive coefficient of 15969.800, implying an increase in the interest rate would mean an increase in consumer spending. This aligns with what was hypothesized.

The p-value for time is 0.187, suggesting insignificance. Time has a coefficient of -1362.917, which implies that as time increases by one, consumer spending decreases by 1362.917. This does not align with what was hypothesized.

Model #2: Cochrane-Orcutt Regression - Regressing Consumer Spending on GDP Deflator

Iteration 0: rho = 0.0000
 Iteration 1: rho = -0.1140
 Iteration 2: rho = -0.1653
 Iteration 3: rho = -0.1870
 Iteration 4: rho = -0.1961
 Iteration 5: rho = -0.1998
 Iteration 6: rho = -0.2013

Cochrane-Orcutt regression

CS	Coefficient	Std. err.	t	P>t	[95% conf. interval]
deflator	4889.249	3504.408	1.400	0.181	-2504.406 12282.900
gdpgrowth	1405.799	2321.749	0.610	0.553	-3492.663 6304.261
interestrate	41230.920	13747.450	3.000	0.008	12226.340 70235.510
time	-268.387	1186.135	-0.230	0.824	-2770.912 2234.138
_inter	5.76e+05	2387475	0.240	0.812	-4460865 5613398
rho	-0.202	0.241	-0.840	0.414	-0.711 0.307

Number of Obs. 18

R-squared 0.7407

F(4, 13) 9.27

Prob > F 0.0009

Durbin-Watson statistic (original) = 2.150880

Durbin-Watson statistic (transformed) = 2.070462

The regression of consumer spending on GDP deflator has an r-squared of 0.7407, implying 74.07% of the variance in the dependent variable in this model is explained by the

independent variables. The F-statistic in this model is 0.0009 indicating overall significance.

The p-value for GDP deflator is 0.181, which indicates an insignificance. GDP deflator has a positive coefficient of 1405.799 which suggests that as GDP deflator increases, so does consumer spending. This reflects what was hypothesized.

The p-value for GDP growth is 0.553, which suggests insignificance. GDP growth has a positive coefficient of 1405.799 implying an increase in GDP growth would mean an increase in consumer spending. This aligns with what was hypothesized.

The p-value for interest rate is 0.008 which suggests extremely good significance. Interest rate has a positive coefficient of 41230.920, implying an increase in the interest rate would mean an increase in consumer spending. This aligns with what was hypothesized.

The p-value for time is 0.824, suggesting insignificance. Time has a coefficient of -268.387, which implies that as time increases by one, consumer spending decreases by 268.387. This does not align with what was hypothesized.

Model #3: Pooled OLS Regression - Regressing Consumer Spending on CPI Trim

Linear regression

CS	Coef.	St.Err.	t-value	p-value	[95% Conf	Interval]	Sig
trim	14735.013	7507.004	1.96	.07	-1365.909	30835.935	*
gdpgrowth	4167.009	1883.984	2.21	.044	126.265	8207.753	**
interestrate	34322.224	13056.412	2.63	.02	6319.006	62325.443	**
time	287.731	1205.305	0.24	.815	-2297.392	2872.854	
Constant	-533379.28	2427875.2	-0.22	.829	-5740653.6	4673895.1	
Mean dependent var		45219.105	SD dependent var			43598.405	
R-squared		0.729	Number of obs			19	
F-test		9.426	Prob > F			0.001	
Akaike crit. (AIC)		444.015	Bayesian crit. (BIC)			448.737	
*** $p<.01$, ** $p<.05$, * $p<.1$							

The regression of consumer spending on CPI trim has an r-squared of 0.729, implying 72.9% of the variance in the dependent variable in this model is explained by the independent variables. The F-statistic in this model is 0.001 indicating overall significance.

The p-value for CPI trim is 0.07, which indicates a good significance level. CPI trim has a positive coefficient of 14735.013 which suggests that as CPI trim increases, so does consumer spending. This is on track with what was hypothesized.

The p-value for GDP growth is 0.044, which suggests a very good significance level. GDP growth has a positive coefficient of 4167.009 implying an increase in GDP growth would mean an increase in consumer spending. This aligns with what was hypothesized.

The p-value for interest rate is 0.02 which suggests a very good significance level. Interest rate has a positive coefficient of 34322.224, implying an increase in the interest rate would mean an increase in consumer spending. This aligns with what was hypothesized.

The p-value for time is 0.815, suggesting insignificance. Time has a coefficient of 287.731, which implies that as time increases by one, consumer spending increases by 287.731. This aligns with what was hypothesized.

Model #4: Pooled OLS Regression - Regressing Consumer Spending on CPI Median
Linear regression

CS	Coef.	St.Err.	t-value	p-value	[95% Conf	Interval]	Sig
median	22604.258	10920.336	2.07	.057	-817.533	46026.049	*
gdpgrowth	3535.288	1832.468	1.93	.074	-394.964	7465.541	*
interestrate	35754.709	12521.668	2.86	.013	8898.403	62611.016	**
time	79.025	1188.748	0.07	.948	-2470.586	2628.636	
Constant	-114061.79	2394410.3	-0.05	.963	-5249561.2	5021437.6	
Mean dependent var		45219.105	SD dependent var		43598.405		
R-squared		0.736	Number of obs		19		
F-test		9.739	Prob > F		0.001		
Akaike crit. (AIC)		443.561	Bayesian crit. (BIC)		448.283		
*** $p<.01$, ** $p<.05$, * $p<.1$							

The regression of consumer spending on CPI median has an r-squared of 0.736, implying 73.6% of the variance in the dependent variable in this model is explained by the independent variables. The F-statistic in this model is 0.001 indicating overall significance.

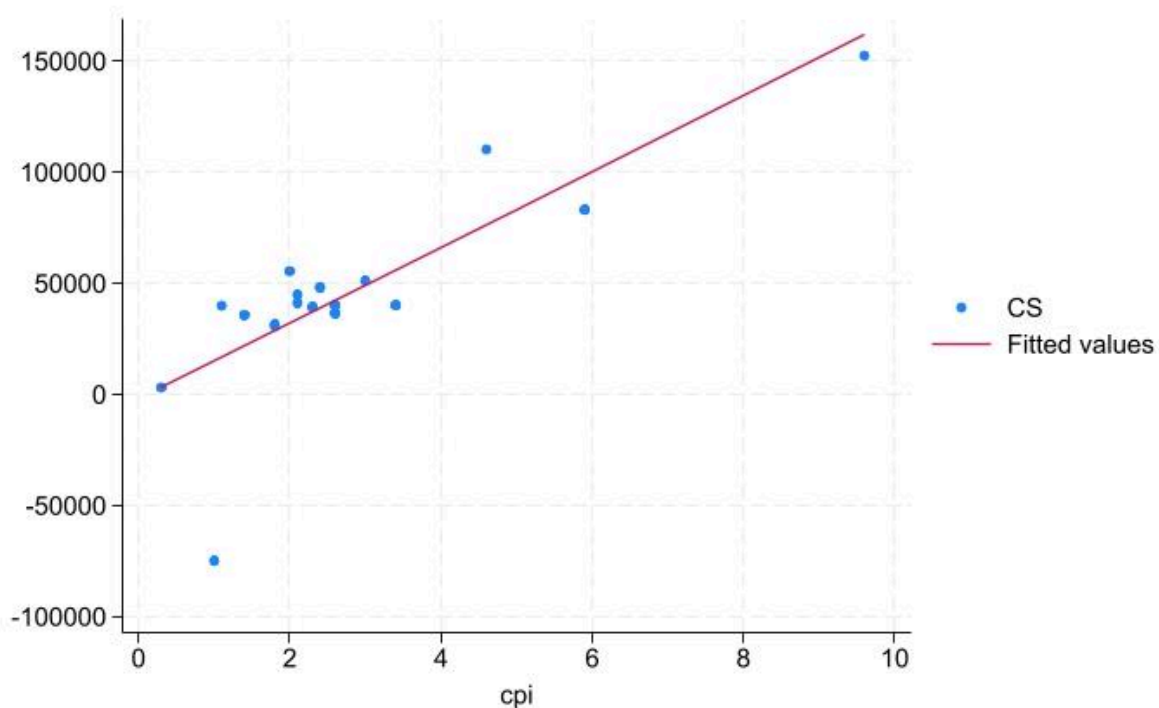
The p-value for CPI trim is 0.057, which indicates a good significance level. CPI trim has a positive coefficient of 22604.258 which suggests that as CPI trim increases, so does consumer spending. This is on track with what was hypothesized.

The p-value for GDP growth is 0.074, which suggests a good significance level. GDP growth has a positive coefficient of 3535.288 implying an increase in GDP growth would mean an increase in consumer spending. This aligns with what was hypothesized.

The p-value for interest rate is 0.013 which suggests a very good significance level. Interest rate has a positive coefficient of 35754.709, implying an increase in the interest rate would mean an increase in consumer spending. This aligns with what was hypothesized.

The p-value for time is 0.963, suggesting insignificance. Time has a coefficient of 79.025, which implies that as time increases by one, consumer spending increases by 79.025. This aligns with what was hypothesized.

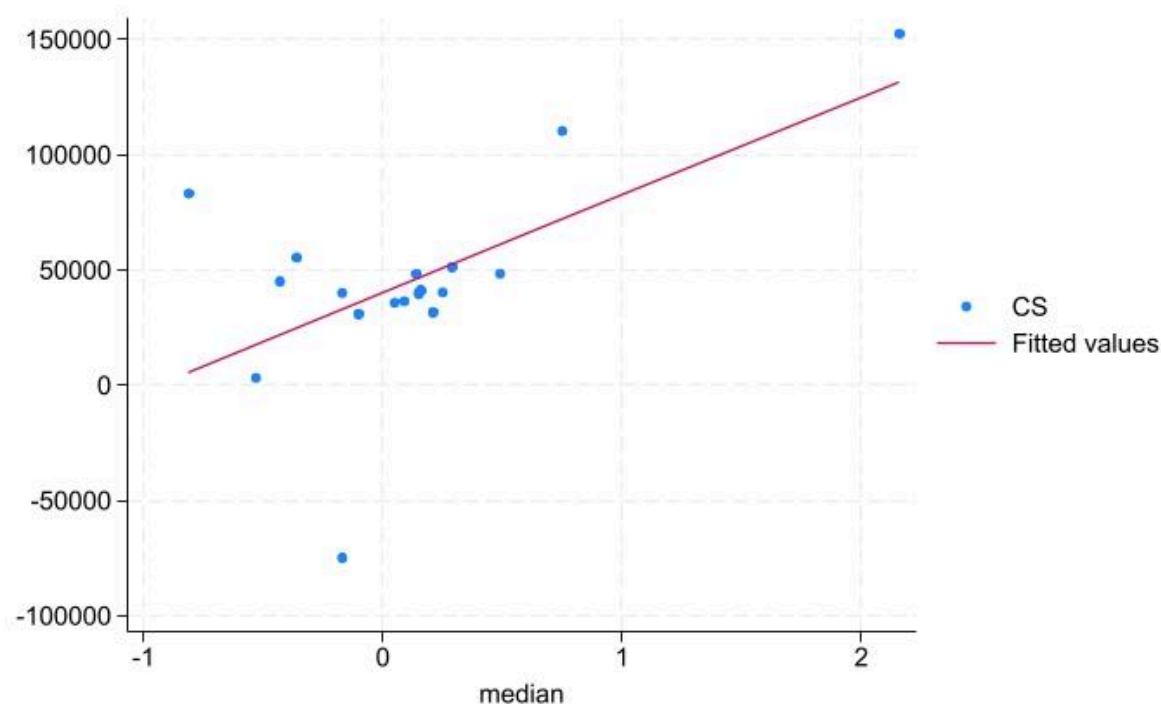
Figure #1 - Twoway Scatterplot between Consumer Spending and CPI



This scatter plot shows the relationship from model #1, which regresses consumer spending on CPI. This is the most efficient and accurate model. As seen above, the relationship

is positive with the dots creating an upward sloping line. This emphasises how as CPI (inflation increases), so does consumer spending. There is one extreme outlier in this scatterplot which is most likely explained by a significant decrease in consumer spending one year, likely near COVID.

Figure #2 - Twoway Scatterplot between Consumer Spending and CPI Median



This scatter plot shows the relationship from model #4, which regresses consumer spending on CPI median. This is an efficient and accurate model. As seen above, the relationship is positive with the dots creating an upward sloping line. This emphasises how as CPI median (inflation increases), so does consumer spending. There is a singular extreme outlier in this scatterplot which is likely best explained by a significant decrease in consumer spending one year, likely near COVID.

Section 5: Limitations

During this research, there was one major limitation which impeded on my analysis. This limitation was positive autocorrelation. At first, I had four OLS regressions with the raw data obtained from various sources. Upon regressing, it became apparent that positive autocorrelation was present. I performed tests such as the durbin watson test to seek if autocorrelation was present. If the durbin-watson statistic is between 0 and 2, this suggests positive autocorrelation. (Analyttica Datalab, 2021) If the statistic is at or around 2, this suggests there is no autocorrelation present. (Analyttica Datalab, 2021) If the statistic is between 2 and 4, this suggests that negative autocorrelation is present. (Analyttica Datalab, 2021) For each of my four models, the d-watson statistic was between 0.222360 and 0.833782, suggesting serious positive autocorrelation. I had to take measures in order to fix this, while simultaneously attempting to maintain the strong results I had obtained. The first measure I took to combat autocorrelation was the Cochrane-Orcutt regression technique. I substituted this method into former OLS regressions that seemed fit. This took place for my CPI and GDP deflator models. As mentioned earlier, the Cochrane-Orcutt regression is a regression technique which “...take[s] serial correlation into account in the error term of linear model[s].” (Lumivero, n.d.-a) The next measure I took was differencing every variable in each model. “Differencing [helps] stabilise the mean of a time series by removing changes in the level of a time series, and therefore eliminating (or reducing) trend and seasonality.” (Athanasopoulos & Hyndman, 2018) The third and final measure I took to account for autocorrelation was adding a time trend variable. Adding a time trend variable “...introduce[s] time-varying coefficients and, thus, approximate unobserved functional structure in economic models...”. (Encyclopedia.com, n.d.) This is extremely useful for my models as it helps account for autocorrelation. I used a time trend variable for all four models in my research. All three of these measures have been implemented in my models to fix for autocorrelation. The

d-watson statistics have improved significantly after using these measures, ranging now from 1.996845 to 2.070462.

Conclusion

In conclusion, this research paper aimed to consider and analyse the impacts of inflation on consumer spending in Canada between the years of 2004 and 2023. This was done by implementing four different models which each, in their own way, measured how consumer spending reacts as inflation increases. These models all regressed on consumer spending, with the main independent variable differing across each model. These independent variables were CPI, CPI mean, CPI trim, and GDP deflator.

The research done in this paper has shown that there is a strong positive relationship between inflation and consumer spending. As inflation increases, so does consumer spending in Canada. This aligns with what was hypothesized. The independent variables show to be significant, with the exception of GDP deflator. CPI was very significant, which was as expected since it is the main indicator of inflation. The positive relationship between inflation and consumer spending was very positive, shown by high coefficients of inflation indicator independent variables in each model.

Generally speaking, the control variables (GDP growth and interest rate) were significant for the models given and therefore suggest a strong relationship with the dependent variable. The interaction variable was not significant, however was still useful to deal with/identify autocorrelation.

This research can be enhanced in the future by using one model with the same variables and more. The model of choice would definitely be model #1 with CPI as the inflation indicator.

This model was proven to be the most efficient, therefore thoroughly examining and adding to just this model would be beneficial for better results and improved research on this important topic dealing with inflation and consumer spending.

The research done in this paper has helped answer the question of what the impact of inflation is on consumer spending in Canada. This should be helpful for future research done on this topic as it will be insightful. This research should also be helpful for policymakers and to predict future patterns of inflation.

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